MIL-P-21251D
8 February 1985
SUPERSEDING
MIL-P-21251C
5 September 1979

MILITARY SPECIFICATION

PUMPING UNITS, SEWAGE, DUPLEX, CENTRIFUGAL, AUTOMATIC, WET-PIT TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic, electric-motor-driven, duplex, centrifugal, wet-pit type, sewage pumps. Capacities for each pump of the duplex unit range from 50 to 1,500 gallons per minute (gpm); total dynamic heads range from 10 to 90 feet.

1.2 Part number. Part numbers shall be assigned as follows: for example see 6.9.

M21251	XXXX	XX	X
Specification part number			
	Gallons per minute (see 3.8.1)		
		Total dynamic head (see 3.8.1)	
			Material designation (see 3.4.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture - and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4630

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MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A120 - Pipe, Steel, Black and Hot-Dipped Zinc Coated (Galvanized) Welded and Seamless, for Ordinary Use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE (HI)

Standards of Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Ave., Cleveland, OH 44107.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The duplex pump unit shall consist of two individual, vertical, submerged-type, nonclogging sewage pumps supported from motor by extended discharge column and pump shaft. Each pump shall be furnished with a vertical electric motor, pump-mounting plate, pump support pipe, discharge pipe, lubrication system, and automatic controls. Options are included in this specification to provide for indoor installation of the pump units and for outside installation where water-tight enclosures for electrical equipment are required.

3.2 Standard commercial product. The sewage pump shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the sewage pumps being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish one sewage pump for first article inspection and approval (see 4.2.1 and 6.2).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.4.1 Material designation. The requirements for each material designation shall be in accordance with the applicable definition listed in the Centrifugal Pump Section of the Hydraulic Institute Standards. Material for pumping unit components and accessories not covered by these designations shall be as specified herein. The pump material designation, when specified (see 6.2), shall be one of the following:

a. All iron pump.
b. Bronze-fitted pump.
c. All bronze pump.
d. Stainless steel-fitted pump.

3.4.2 Dissimilar metals. Contact between dissimilar metals likely to cause deterioriation of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, insulation, organic coatings, gaskets, or other suitable means.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. Pump units and accessories shall be designed to permit accessibility for maintenance and service in the field, and to prevent conditions which may be hazardous to personnel. Ample protection against electrical shock shall be provided. Screw threads and pipe threads shall conform to the requirements of FED-STD-H28.

3.7 Construction. The plan dimensions of the sump cover plate and the length of the pump from the sump cover plate to the bottom of the pump section inlet shall be by the requirements of the procuring activity. These requirements may be indicated in the invitation for bids by any one of the following options:

a. Depth and plan dimensions of Government furnished sump, for which pump is intended, as specified (see 6.2).
b. The manufacturer's standard sump tank is to be furnished with dimensions of pump and tank to be in accordance with manufacturer's recommendations for the specified operating characteristics of the pump (see 3.9.3).
c. If a sump tank is to be furnished, dimensions of the tank are to be as specified by the procuring activity (see 3.9.3).

In the event the length of pump is by (a) or (c), the dimensions of the pump from the sump cover plate to the bottom of the suction inlet shall be such as to provide a clearance of not less than one discharge pipe diameter nor more than 18 inches from the floor of the sump or sump tank.

3.7.1 Pump casing. The volute and discharge nozzle of the pump casing shall be cast in one piece. The casing shall be constructed with a bolted plate to permit inspection and removal of the impeller. The casing shall be designed to withstand a hydrostatic pressure of not less than 2.0 times the maximum rated head of the pump.

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3.7.2 Impeller. The impeller shall be nonclogging type with blunt entering edges and other design features required to provide maximum freedom from clogging when sewage containing rags and stringy material is handled. The impeller shall be balanced as required to prevent harmful vibrations during pumping operations. Unless otherwise specified (see 6.2), the impeller shall be of the totally enclosed type.

3.7.3 Shaft. Pump shafts shall be high carbon or alloy steel, unless stainless steel or nickel-copper alloy shafts are specified (see 6.2). The mechanical properties and diameter of the shaft shall be such as to insure that neither whip, deflection, nor vibration will be of sufficient magnitude to impose greater than design loads on the specified shaft bearings under normal operating conditions. Means shall be provided on the part of the shaft extending above the sump cover plate for adjustment of the clearance between the impeller and the inner surfaces of the volute section.

3.7.4 Bearings. One or more radial ball bearings designed and located to carry mechanical and hydraulic thrust loads of rotating components shall be furnished. The thrust bearings shall be located above the pump-mounting plate and fitted with a moisture-proof seal. Intermediate shaft bearings shall be of the sleeve type, so located that the center distance between any two bearings on the shaft does not exceed 4 feet 6 inches for pumps operating between 1,700 and 1,800 revolutions per minute (rpm) and does not exceed 5 feet for pumps operating at 1,200 rpm or less. A sleeve-type bearing, located near the lower extremity of the shaft shall also be furnished. Means shall be provided to prevent the pump fluid from entering the lower bearing. Such means may include a suitable seal or a system wherein a partial vacuum developed below the bearing by the impeller rotation, induces a positive flow of lubricant into the bearing. Sleeve type bearings shall be heavy-duty bronze or bronze-backed, babbit-lined. Appropriate piping and fittings shall be provided to permit individual lubrication of the intermediate and lower bearings from above the sump cover plate.

3.7.5 Support pipe. A support pipe concentric with the pump shaft shall connect the pump to the sump cover plate. The support pipe flanges shall be suitably machined or doweled to insure proper alinement of the pump and shaft when the pipe is assembled.

3.7.6 Discharge pipe. A discharge pipe running from the pump discharge outlet through the sump cover plate shall be furnished as an integral part of the pump unit. The pipe shall be black steel or wrought iron with wall thickness not less than that specified in ASTM A120 for schedule 40 pipe. The discharge pipe shall be gastight through the sump cover plate. Unless otherwise specified (see 6.2), the discharge end of the pipe shall terminate in a screwed or flanged connection in accordance with the manufacturer's standard practice.

3.7.7 Sump cover plate. The sump cover plate shall be cast iron or steel and shall be adequate strength to support without visible distortion both the pump(s) and maintenance personnel required for normal service. All openings through the sump cover plate shall be gasketed or otherwise sealed to prevent the escape of sump gases except through the vent. A standard 4-inch threaded outlet shall be provided to permit installation of a vent pipe.

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3.8 Performance.

3.8.1 Capacity. Each pump of the duplex pump unit shall deliver at rated speed, not less than the specified gpm against the specified discharge heads (see 6.2 and 6.3), while the liquid level is not more than 1 foot above the datum elevation of the pump. The datum elevation shall be the level of the entrance eye of the impeller. The discharge head includes the friction head of the system piping external to the pump unit and the static head measured from a point of reference on the sump to the highest point in the system. The manufacturer shall make allowances for the difference in elevation between the indicated point of reference and the datum elevation, and for friction losses in the discharge piping. Ratings shall be based on pumping clear fresh water at a temperature of 68 degrees Fahrenheit (°F).

3.9 Components and accessories.

3.9.1 Electric motors. Electric motors shall be of the vertical type and shall be rated and constructed in accordance with the applicable provisions of NEMA MG 1. Motors shall be general purposed, continuous-duty with a temperature-rise rating of 40° Celsius (C) (or a rating of 40°C ambient). The synchronous speed rating of motors shall not exceed 1,800 rpm for 60 Hertz (Hz) operation and 1,500 rpm for 50 Hz operation. Motors shall be furnished with drip-proof or NEMA type II, weather-protected enclosures as specified (see 6.2). Voltage, phase, and frequency ratings shall be as specified (see 6.2). Provisions shall be made within the motor to prevent overloading at any point on the pump performance curve in accordance with NFPA No. 70. The horsepower (hp) rating of each pump motor of the duplex pump unit shall be such as to insure conformance to the following requirements:

a. The motor hp rating shall not be exceeded when the pump is delivering the specified volume of clear fresh water at a temperature of 68°F against the total dynamic head of the intended installation.
b. With the pump delivering the specified capacity of clear fresh water against the total dynamic head of the intended installation, the hp obtained by multiplying the rated hp of the motor by the service factor shall not be exceeded if the static head is decreased by 5 feet, as would occur if the fluid level in the sump rises 6 feet above the datum elevation of the pump (see 6.3).

3.9.1.1 Motor shafts. Unless otherwise specified (see 6.2), motors may be furnished with either solid shafts or hollow shafts, in accordance with the manufacturer's standard practice. Solid-shaft motors shall be furnished with a thrust bearing capable of carrying the weight of the rotor. Solid shafts shall be connected to the pump shaft by a flexible coupling. Solid shaft motors shall be mounted on a motor-support pedestal of sufficient height to provide ascessibility to the coupling, pump shaft thrust bearing, and the pump shaft height adjustment. Hollow shaft motors shall be furnished with a thrust-bearing designed for carrying the combined mechanical and hydraulic thrust loads of the pump as well as the weight of the rotor. Thrust bearings for both solid and hollow shaft motors shall conform to the applicable requirements of 3.7.4.

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3.9.2 Electrical controls. Enclosures for control switches shall conform to the requirements of NEMA ICS 1 and shall be NEMA type 1, general purpose of NEMA type 4, water-tight, as specified (see 6.2). Float controls and high-water alarm devices shall be mounted on the sump cover plate. Other electrical components such as magnetic starters, disconnect switches, circuit breakers, and control panels shall be furnished for wall mounting above the floor.

3.9.2.1 Liquid-level control. The duplex pump unit shall be furnished with the electrical and mechanical devices required to provide automatic operation of the pump unit when the liquid level in the sump rises to a predetermined level. The controls shall automatically transfer the operating cycle from one pump to the other and shall operate both pumps simultaneously whenever the inflow to the sump exceeds the capacity of the operating pump. means of adjustment, such as float rod stops, shall be provided to allow for variations in the fluid level at which the pump unit is automatically started. All parts of the fluid level, sensing-mechanism in contact with the fluid shall be bronze, brass, or material of equivalent resistance to the corrosive effects of sewage.

3.9.2.2 High-water alarm. When specified (see 6.2), a high-water alarm switch complete with actuating-mechanism shall be provided for operation on an electrical circuit other than the motor circuit. The switch shall be designed to operate an alarm bell or warning light whenever a predetermined high-water level is reached in the sump because of either pump failure or a fluid inflow that exceeds the combined capacity of both pumps.

3.9.3 Sump tank. When specified (see 6.2), a cast iron or steel sump tank shall be furnished with the duplex pump unit. Unless otherwise specified (see 6.2), the dimensions of the tank shall be as recommended by the manufacturer for use with the pumps furnished under this specification. A standard opening for connection to the sewage inflow pipe shall be furnished in the specified size and location with respect to the top of the tank (see 6.2).

3.10 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 pound-force per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.11 Treatment and painting. Unless otherwise specified (see 6.2), the pumping unit shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the pumping unit other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

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3.12 Wiring. All wiring shall be in accordance with NFPA No. 70.

3.13 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.
c. One percent of copper 8-quinolinolate (by weight based on the nonvolatile content of the varnish) may, at the contractor's option, be substituted for the salicylanilide specified in MIL-V-173, provided that the varnish conforms to MIL-V-173 in all other aspects.

3.14 Identification marking. Identification shall be permanently and legibly marked directly on the side of the pump or on a corrosion-resisting metal plate securely attached to the pump at the source of manufacturer. Identification shall include the manufacturer's model and serial number, pump type, capacity and head, motor hp, and name and trademark to be readily identifiable to the manufacturer.

3.15 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall

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be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. Field inspection (see 4.2.3).

4.2.1 First article inspection. The first article inspection shall be performed on one pump unit when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.2.3 Field inspection. When specified (see 6.2), field inspection shall be performed on the pump unit(s) at the site of installation. This inspection shall consist of all tests deemed necessary by the procuring activity to verify compliance with the requirements of this specification. Field inspection shall be performed either by the Government, in which case the contractor shall have the privilege of representation, or by the contractor,

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as specified (see 6.2). In either case, deficiencies revealed during such tests shall be corrected at the contractor's expense and any required retesting shall be at the contractor's expense.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one pump unit. All pump units offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 4.0 percent defective.

4.4 Examination. The first article, when required, and each sample selected in accordance with 4.3.1 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table I. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE I. Classification of defects.

Classification	Defects	Requirement Paragraph
Critical	None defined.	
Major:		
101	Pumping unit not complete.	1.1 & 3.1
102	Material not as specified.	3.4,3.4.1 & 3.4.2
103	Component parts not new or interchangeable.	3.5
104	Pump units not accessible for service.	3.6
105	Hazardous to personnel.	3.6
106	Screw & pipe threads not as specified.	3.6
107	Dimensions not as specified.	3.7
108	Pump casing not as specified.	3.7.1
109	Impeller not proper type and not as specified.	3.7.2
110	Pump shaft not as specified.	3.7.3
111	Bearings not as specified.	3.7.4
112	Support pipe not properly aligned.	3.7.5
113	Discharge pipe not as specified.	3.7.6
114	Sump cover plate not as specified.	3.7.7
115	Electric motor not of the type specified or rating required.	3.9.1

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TABLE I. Classification of defects. cont'd

116	Motor shafts not as specified.	3.9.1.1
117	Electrical controls not furnished with specified type of enclosure.	3.9.2
118	Liquid level control not as specified.	3.9.2.1
119	High water alarm not furnished when specified.	3.9.2.2
120	Sump tank when furnished, not of specified size.	3.9.3
121	Pump not lubricated and lubrication fittings not as specified.	3.10
122	Improper wiring.	3.12
123	Not treated for fungus resistance, when required.	3.13
Minor:		
201	Part number omitted or not correct.	1.2
202	Treatment & painting not as specified.	3.11
203	Equipment not marked as specified or not furnished with instruction plates when required.	3.14 & 3.15
204	Poor workmanship.	3.16

4.5 Tests.

4.5.1 Function test. The duplex pump unit shall be operated as required to verify that all mechanical and electrical components will function in a manner to insure compliance with this specification when the equipment is installed in its intended location. Manual activation of float-actuated controls may be employed to simulate fluctuation of the liquid level.

4.5.2 Performance test. Each pump of the duplex pump units shall be tested to verify compliance with the hyrdaulic performance requirements of 3.8.1 and the electrical performance requirements of 3.9.1, 3.9.2.1, and, when applicable, 3.9.2.2. Test methods and procedures shall be in accordance with the Test Code, Centrifugal Pump Section of the Standards of the Hydraulic Institute. During the test run a characteristics head-capacity curve shall be plotted within the limits of the operating range imposed by the hp rating of the motor. Brake hp input to the pump at the specified discharge rate and dynamic head shall also be checked and recorded duing the test run. If the length of the pump unit furnished under this specification precludes the use of the manufacturer's available testing facility, a shorter support pipe, pump shaft, discharge pipe, and float mechanisms may be substituted for testing purposes, providing other requirements for the preproduction sample are met. Failure of the pump unit to meet any of the performance requirements of this specification shall be cause for rejection.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-P-10603. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

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5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-10603 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. This specification covers pumps intended for use in pumping raw sewage that carries solids from a sump tank or tank into a sewer at higher level.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and aproval (see 3.3, 4.2.1, and 6.7).
c. Material designation for pump (see 3.4.1).
d. Depth and plan dimensions of sump, when sump is to be Government furnished (see 3.7).
e. When semi-open type impellers are required (see 3.7.2).
f. When stainless steel or nickel-copper alloy pump shafts are required (see 3.7.3).
g. When either a flanged or screwed discharge connection is specifically required (see 3.7.6).
h. Capacity of each pump of duplex unit in gpm and discharge head in feet of intended installation. Also location point of reference on sump used to determine discharge head. (see 3.8.1).
i. Type of motor enclosure required (see 3.9.1).
j. Voltage, phase, and frequency of electric power at site of installation (see 3.9.1).
k. When either solid shaft of hollow shaft motors are specifically required (see 3.9.1.1).
l. Type of enclosure required for electrical controls (see 3.9.2).
m. When a high water alarm switch is required (see 3.9.2.2).
n. When sump tank is required and when drawings shall be furnished prior to award of contract (see 3.9.3).
o. Dimensions of sump tank required if manufacturer's standard tank is not acceptable (see 3.9.3).
p. Location and size of sewage connection when sump tank is furnished (see 3.9.3).
q. Means for lubrication other than the manufacturer's standard practice (see 3.10).
r. Treatment and painting, if different (see 3.11).
s. When fungus control of electrical components is required (see 3.13).
t. When first article tests shall be other than specified (see 4.2.1).
u. When field tests are required, location of tests, and whether tests are to be performed by the contractor or the Government (see 4.2.3).
v. When preproduction pack inspections are required and time frame required for submission (see 4.2.1 and 4.6).
w. Level of preservation and level of packing required (see 5.1).

6.3 Capacity. The design of nonclogging impellers is such that, a capacity increases with a reduction in head, the hp input requirements of the pump will also increase. Therefore, the total head should be determined as

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accurately as possible to eliminate possible overloading of the motor by specifying a head that is greater that the actual heads against which the pump will be required to operate.

6.4 Sump dimensions. Most contractors can furnish standard pumps for sump depths up to 10 feet in increments of 6 inches. A 20-foot maximum length is offered by some manufacturers. Standard sump tanks are available with diameters or square length ranging form 42 inches up to 108 inches in increments of 1 foot.

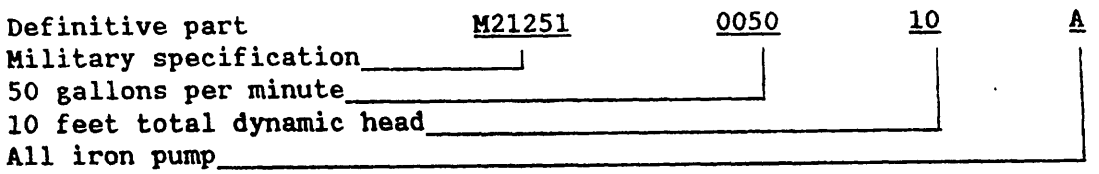
6.5 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.6 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Definitive part number, example. The Military specification number and the specification part number combine to form the definite part number.

Example:



Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 4630-0012)

Review Activities:

Army - CE, ER
Navy - MS
DLA - CS

User Activities:

Navy - MC, CG

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-21251D	PUMPING UNITS, SEWAGE, DUPLEX, CENTRIFUGAL, AUTOMATIC, WET-PIT TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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